

DIRNET-OOD: Domain-Agnostic Incremental Routing Network with Out-of-Distribution Residual Prior

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Abstract—Domain-agnostic incremental sound classification requires a model to sequentially learn new acoustic domains while preserving previously acquired knowledge and performing inference without access to test-time domain labels. This setting is challenging because only data from the current domain are available during incremental learning, whereas evaluation samples may originate from any previously learned domain. Existing domain-specific classifiers often rely on known domain labels or unstable confidence-based routing, limiting their robustness in domain-agnostic inference. This paper proposes DIRNET-OOD (Domain-Agnostic Incremental Routing Network with Out-of-Distribution Residual Prior), a CNN14-based incremental learning framework with prototype residual routing for domain-agnostic inference. The proposed method performs prototype residual routing by constructing class-wise prototypes for D2 and D3 in a learned routing space and estimating domain responsibility using probability-weighted prototype distances. Since D1 development data cannot be accessed during later incremental stages, the D1 branch is incorporated through an OOD residual prior instead of explicit prototypes. In addition, adaptive D2/D3 pairwise calibration refines the relative contributions of the D2 and D3 branches before soft probability fusion, improving routing stability when their prototype distances are similar. Experiments on the DCASE 2026 Task 7 benchmark show that DIRNET-OOD improves the macro-average accuracy on D2/D3 evaluation samples from 52.6% to 74.89%, corresponding to a 22.29 percentage-point improvement over the official baseline. These results demonstrate that class-wise prototype routing, combined with an OOD residual prior and adaptive pairwise calibration, is effective for domain-agnostic incremental sound classification.

Index Terms—domain-agnostic incremental learning, out-of-distribution, prototype routing, soft probability mixture

1. INTRODUCTION

Domain-incremental learning has become an important problem in acoustic scene and sound event classification, where a model must adapt to newly observed domains while retaining knowledge from previously learned domains. In the DCASE 2026 Task 7 setting, acoustic domains are introduced sequentially in the order $D1 \rightarrow D2 \rightarrow D3$, while the target sound classes remain fixed. During evaluation, however, test samples may originate from any learned domain and the corresponding domain labels are not provided. Therefore, the model must not

only classify the input sound but also determine how to utilize domain-specific knowledge under unknown-domain conditions.

This problem is challenging for two main reasons. First, acoustic characteristics can vary significantly across domains due to differences in recording conditions, background noise, and domain-specific signal distributions. Second, only the current-domain data are available during incremental learning, which makes the model vulnerable to catastrophic forgetting and limits direct access to previous-domain distributions. The official baseline [1] addresses this problem using domain-specific batch normalization and entropy-based path selection. However, confidence- or entropy-based hard routing can be unstable when the model is overconfident on samples from an unknown or ambiguous domain.

To address these limitations, this paper proposes DIRNET-OOD, a Domain-Agnostic Incremental Routing Network with an out-of-distribution residual prior. The proposed method maintains domain-specific classification paths based on a CNN14 backbone [2] and performs soft domain fusion at inference time. Instead of selecting a single domain path using confidence or entropy, DIRNET-OOD estimates domain responsibility using class-wise prototype [3], [4] distances in a learned routing space. Since D1 development data are unavailable in later incremental stages, the D1 branch is incorporated through an OOD residual prior [5] rather than explicit D1 prototypes. In addition, adaptive D2/D3 pairwise calibration is introduced to refine the relative contribution of the D2 and D3 branches when their prototype distances are similar.

The main contributions of this paper are summarized as follows:

- We propose DIRNET-OOD, a prototype residual routing framework for domain-agnostic incremental sound classification without test-time domain labels.
- We introduce probability-weighted class-wise prototype routing for D2 and D3, enabling class-conditional modeling of domain shift in a learned routing space.
- We incorporate the unavailable D1 domain using an OOD residual prior, allowing the D1 branch to contribute during inference without storing D1 development data.
- We apply adaptive D2/D3 pairwise calibration to improve routing stability when D2 and D3 prototype distances are

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similar.

The remainder of this paper is organized as follows. Section 2 reviews related work on incremental audio classification and routing-based domain adaptation. Section 3 describes the proposed DIRNET-OOD framework. Section 4 presents the experimental setup, Section 5 reports the results and discussion, and Section 6 concludes the paper.

2. RELATED WORK

2.1. Incremental learning for audio classification

Incremental learning adapts a model to new data while preserving previously learned knowledge. In audio classification, this is challenging because recording devices, background noise, and acoustic conditions can change across domains. Prior work has addressed catastrophic forgetting [6] through knowledge distillation [7], replay, domain-specific normalization, and parameter isolation. DCASE 2026 Task 7 is particularly challenging because previous-domain data are unavailable during incremental learning and domain labels are not provided during evaluation.

The official baseline uses a CNN-based classifier with domain-specific components and entropy-based hard path selection. However, entropy does not always provide a reliable estimate of domain membership. This motivates our use of prototype-based domain responsibility estimation and soft fusion across domain-specific branches.

2.2. Prototype-based routing and out-of-distribution modeling

Prototype-based routing represents classes or domains using reference feature vectors and assigns inputs based on distances in an embedding space. This is useful for domain-agnostic inference because it provides an explicit similarity measure to stored domain references.

In DIRNET-OOD, D2 and D3 are represented by class-wise prototypes, while D1 is handled using a residual prior because D1 development data are unavailable. This design preserves a limited contribution from the D1 branch while relying on D2/D3 prototype distances to select or fuse later-domain branches.

3. PROPOSED METHOD

This section describes the proposed DIRNET-OOD framework. The overall architecture is shown in Fig. 1. The method consists of a CNN14-based domain-specific classification architecture, sequential adaptation for D2 and D3, class-wise prototype routing for D2/D3, an OOD residual prior for D1, and adaptive D2/D3 pairwise calibration.

3.1. Overall Architecture

DIRNET-OOD uses a CNN14/PANNS-style backbone to extract a 2048-dimensional clip-level embedding from a log-mel spectrogram. Each CNN14 branch consists of six convolutional blocks, where each block contains Conv(3×3), batch normalization, ReLU activation, another Conv(3×3), batch normalization, ReLU activation, and average pooling. The extracted embedding is passed to a domain-specific classifier head for 10-class sound classification.

The framework maintains separate D1, D2, and D3 branches. The D1 branch is initialized from the provided D1 checkpoint. The D2 branch is initialized from D1 and adapted using D2 data, while the D3 branch is initialized from the learned D2 branch and adapted using D3 data. During evaluation, all domain branches are evaluated in parallel. Instead of selecting a single branch, DIRNET-OOD estimates domain-specific routing weights and obtains the final prediction through soft probability fusion.

3.2. Incremental Learning Procedure

The incremental learning process follows the DCASE 2026 Task 7 protocol. At the D2 stage, the model starts from the provided D1 checkpoint and is trained using only D2 development data. The D1 branch is preserved to maintain previously learned knowledge, while the D2 branch is adapted to the new acoustic domain. After D2 training, class-wise D2 prototypes are recomputed using the trained D2 branch with data augmentation disabled.

At the D3 stage, the model is initialized from the learned D2 branch and trained using only D3 development data. After D3 training, class-wise prototypes for both D2 and D3 are recomputed using clean forward passes. This prototype recomputation step is important because prototypes computed during augmented training may not provide stable routing references. By recomputing prototypes without augmentation, the routing space better reflects the evaluation-time feature distribution.

3.3. Prototype Residual Routing

DIRNET-OOD performs prototype-based routing in a 128-dimensional normalized routing space derived from the 2048-dimensional CNN14 embedding. A routing projection head maps the clip-level embedding to the routing feature space. Let x denote an input audio clip, $r \in \mathbb{R}^{128}$ denote the normalized routing feature, and $p_{d,c}$ denote the prototype of class c in domain $d \in \{D2, D3\}$. Let $P_d(c|x)$ be the class posterior probability predicted by domain branch d .

For each domain $d \in \{D2, D3\}$, the prototype-based distance is computed as a probability-weighted average of cosine distances to the class-wise prototypes:

$$\zeta_d(x) = \sum_{c=1}^C P_d(c|x) \frac{1 - r \cdot p_{d,c}}{\|r\| \|p_{d,c}\|}. \quad (1)$$

This formulation differs from hard class-based prototype routing because all class prototypes can contribute to the routing decision according to the predicted class posterior. When the classifier output is uncertain, the distance is influenced by multiple candidate classes, which provides a smoother and more reliable estimate of domain responsibility.

The D2 and D3 routing scores are obtained from the prototype distances as

$$s_d = \exp(-\zeta_d/T), \quad d \in \{D2, D3\}, \quad (2)$$

where T controls the sharpness of prototype-distance-based routing [8]. The value is set to $T = 0.15$ in all experiments.

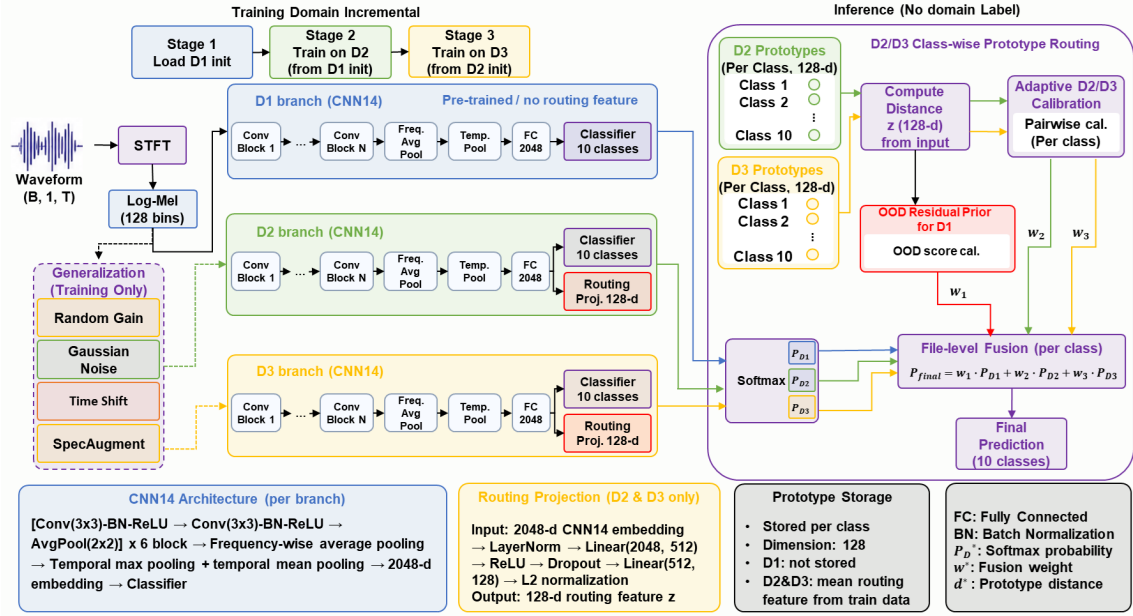


Fig. 1: DIRNET-OOD architecture.

Since D1 development data are unavailable, D1 prototypes cannot be explicitly computed. Therefore, the D1 branch is incorporated using a fixed D1 residual prior:

$$s_{D1} = \lambda, \quad (3)$$

where λ controls the prior contribution of the D1 branch. The value is set to $\lambda = 0.70$ based on the local D2/D3 development split. Since D1 labels are unavailable, λ was not optimized using D1 accuracy. Instead, it was selected as a routing hyperparameter by comparing D2/D3 macro-average accuracy across candidate values. A smaller value, $\lambda = 0.35$, was also evaluated but resulted in lower D2/D3 macro-average accuracy. Thus, $\lambda = 0.70$ is used as a residual routing prior rather than a calibrated D1 posterior probability.

The initial routing weights are then normalized over the three branches:

$$w_d = \frac{s_d}{s_{D1} + s_{D2} + s_{D3}}, \quad d \in \{D1, D2, D3\}. \quad (4)$$

If an input sample is close to either D2 or D3 prototypes, the corresponding prototype score increases and the D1 contribution decreases after normalization. Conversely, if the sample is far from both D2 and D3 prototypes, the D2/D3 scores decrease and the relative contribution of the D1 residual prior increases. In this way, the D1 branch can contribute to prediction even without explicit D1 prototype storage.

3.4. Adaptive D2/D3 Pairwise Calibration

When the D2 and D3 prototype distances are highly similar, the initial softmax-based routing weights can assign nearly equal responsibility to the two branches. This may be problematic when small but meaningful distance differences should determine the more appropriate branch. To improve D2/D3 discrimination, DIRNET-OOD applies adaptive pairwise calibration [9], [10] after the initial routing weights are computed.

First, the total D2+D3 probability mass is defined as

$$m_{D2+D3} = w_{D2} + w_{D3}. \quad (5)$$

The pairwise distance gap between D2 and D3 is computed as

$$\Delta_{D2,D3} = |\zeta_{D2} - \zeta_{D3}|. \quad (6)$$

Given a target pairwise confidence η , the pairwise temperature is adaptively estimated from the median of the finite D2/D3 distance gaps:

$$T_{D2,D3} = \frac{\text{median}(\Delta_{D2,D3})}{\log(\eta/(1-\eta))}. \quad (7)$$

The target confidence is set to $\eta = 0.8$, encouraging the closer domain to receive approximately 80% of the pairwise responsibility for a typical D2/D3 distance gap. The temperature is then clipped to a predefined range:

$$T_{D2,D3} \leftarrow \text{clip}(T_{D2,D3}, T_{\min}, T_{\max}). \quad (8)$$

The bounds are set to $T_{\min} = 10^{-6}$ and $T_{\max} = 0.05$, respectively. The lower bound prevents numerical instability caused by an extremely small temperature, whereas the upper bound avoids overly flat D2/D3 calibration and preserves the effect of meaningful distance differences.

The relative D2:D3 contribution is then recalibrated using a pairwise softmax:

$$q_d = \frac{e^{(-\zeta_d/T_{D2,D3})}}{e^{(-\zeta_{D2}/T_{D2,D3})} + e^{(-\zeta_{D3}/T_{D2,D3})}}, \quad d \in \{D2, D3\}. \quad (9)$$

Finally, the calibrated D2/D3 routing weights are obtained as

$$\hat{w}_{D2} = m_{D2+D3} q_{D2}, \quad \hat{w}_{D3} = m_{D2+D3} q_{D3}. \quad (10)$$

This procedure only redistributes the relative probability mass between D2 and D3. The D1 routing weight remains unchanged, which preserves the role of the OOD residual prior while improving discrimination between the two prototype-based domains. We additionally use a lightweight D2/D3 router to refine the relative D2:D3 responsibility. Its output is blended with the calibrated prototype-based weights using a blending

factor of 0.2, so prototype-based routing remains the dominant routing signal.

3.5. Prototype Re-computation

Prototype quality directly affects routing reliability. Therefore, after each incremental training stage, prototypes are recomputed using clean forward passes with data augmentation disabled. For each domain $d \in \{D2, D3\}$ and class c , the prototype is computed as the mean of normalized routing features belonging to that class:

$$p_{d,c} = \frac{1}{N_{d,c}} \sum_{i: y_i=c} r_i, \quad (11)$$

where $N_{d,c}$ is the number of samples of class c in domain d and r_i is the normalized routing feature of the i -th sample.

The recomputed prototypes are stored and used only for inference-time routing. No D1 prototype is stored because D1 development data are not available. This design keeps the proposed method consistent with the Task 7 incremental learning constraint while enabling class-conditional routing for the available D2 and D3 domains.

4. EXPERIMENT SETUP

Experiments follow the DCASE 2026 Task 7 protocol. The model starts from the provided D1 checkpoint and sequentially learns D2 and D3 using only the current-domain development data. The final system is evaluated in a domain-agnostic setting, where test-time domain labels are unavailable.

Input audio is processed as a 32 kHz mono waveform and converted into log-mel spectrograms using 64 mel bands, a 1024-sample window, and a 320-sample hop size, following the official baseline. No external data, additional labels, or pretrained embedding models are used. The classification branches are trained with cross-entropy loss and lightweight routing-space auxiliary losses for prototype compactness, D2/D3 prototype separation, and D2/D3 router training.

Training augmentation includes random gain, Gaussian noise, time shift, and SpecAugment [11]. Augmentation is disabled during prototype re-computation to obtain stable evaluation-time routing references. For file-level evaluation, segment-level branch probabilities and routing weights are aggregated over each audio file, and the final prediction is obtained by hybrid file-level probability fusion.

Since D1 development labels are unavailable, model selection and routing calibration are performed using the local D2/D3 development split. The official-style macro-average accuracy is used as the main metric, where class-wise accuracies are averaged within each domain and then averaged across D2 and D3.

5. RESULTS AND DISCUSSION

Table 1 shows the local development macro accuracy and ablation results at the D3 stage. Since D1 development labels are unavailable, only D2 and D3 accuracies are reported. DIRNET-OOD improves the D2/D3 average accuracy from 52.6% to 74.89%, outperforming the official baseline by 22.29 percentage points.

The ablation results show that all proposed components contribute to the final performance. Removing class-wise

Table 1: Local development macro accuracy (%) and ablation results at the D3 stage. D1 accuracy is not reported because development labels are unavailable.

Model	D2	D3	Avg.
Baseline	59.0	46.1	52.6
DIRNET-OOD (full)	79.74	70.04	74.89
w/o class-wise prototype routing	63.34	67.93	65.64
w/o probability-weighted distance	68.21	71.70	69.95
w/o D1 residual prior	69.35	72.50	70.92
w/o adaptive D2/D3 calibration	70.36	72.21	71.28
w/o soft probability fusion	69.22	73.10	71.16
w/o prototype re-computation	61.10	72.76	66.93

prototype routing causes the largest degradation, reducing the average accuracy to 65.64%, which confirms the importance of class-conditional domain prototypes. Prototype re-computation is also important, as disabling it decreases the average accuracy to 66.93%. The other ablations also reduce performance, indicating that probability-weighted distances, the D1 residual prior, adaptive D2/D3 calibration, and soft probability fusion provide complementary gains.

Table 2: Official evaluation macro accuracy (%) at the D3 stage.

Model	D1	D2	D3	Avg.
Baseline	89.15	49.72	42.18	60.35
DIRNET-OOD (full)	88.82	78.62	63.32	76.92

Table 2 reports the official evaluation results. DIRNET-OOD improves the overall macro-average accuracy from 60.35% to 76.92%. The improvement mainly comes from D2 and D3, where the accuracies increase from 49.72% to 78.62% and from 42.18% to 63.32%, respectively. Meanwhile, D1 accuracy remains comparable to the baseline, changing only from 89.15% to 88.82%. These results indicate that DIRNET-OOD improves adaptation to later domains while largely preserving previously learned D1 knowledge.

Overall, the local ablation and official evaluation results demonstrate that the gain of DIRNET-OOD comes from the combined effect of class-wise prototype routing, probability-weighted prototype distance, the D1 residual prior, adaptive calibration, soft fusion, and prototype re-computation.

6. CONCLUSION

This paper presented DIRNET-OOD, a prototype residual routing framework for domain-agnostic incremental sound classification. The proposed method estimates domain responsibility using probability-weighted class-wise D2/D3 prototype distances and incorporates the unavailable D1 domain through an OOD residual prior. Adaptive D2/D3 pairwise calibration further improves routing stability by refining the relative contribution of D2 and D3 before soft probability fusion.

Experiments on the DCASE 2026 Task 7 benchmark show that DIRNET-OOD improves the D2/D3 macro-average accuracy from 52.5% to 74.89%, corresponding to a 22.39 percentage-point improvement over the official baseline. These results demonstrate that the combination of class-wise prototype routing, an OOD residual prior, and adaptive pairwise calibration is effective for domain-agnostic incremental sound classification. Future work will investigate adaptive prototype learning and uncertainty-aware routing to further improve generalization to unseen acoustic domains.

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